

AWS Global Infrastructure

AWS Regions

- An **AWS Region** is a geographical area.
- **Data replication** across Regions is controlled by you.
- **Communication** between Regions uses AWS backbone network infrastructure.
- Each Region provides full redundancy and connectivity to the network.
- A Region typically consists of two or more **Availability Zones**.



Example: London Region



1. AWS Region

- **Definition** : An AWS Region is a physical location around the world where AWS operates multiple data centers.
- **Purpose** : Regions allow users to choose where their resources (e.g., EC2 instances, S3 buckets, databases) are located geographically. This choice can be driven by factors such as latency, compliance, or legal requirements.
- **Characteristics** :
 - Each Region is completely independent and isolated from other Regions.
 - A Region typically consists of two or more Availability Zones.
 - Data and services within a Region are interconnected with low-latency networks.
 - Examples of Regions include **us-east-1** (N. Virginia), **eu-west-1** (Ireland), and **ap-southeast-2** (Sydney).
- **Use Case** : If you need to host your application close to your users for better performance or comply with regional regulations, you can select an appropriate Region.

Selecting a Region

Determine the right Region for your services, applications, and data based on these factors



Data governance, legal requirements



Proximity to customers (latency)



Services available within the Region



Costs (vary by Region)

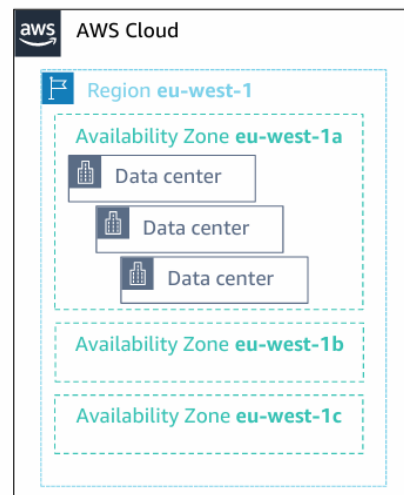


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Availability Zones

- Each **Region** has multiple Availability Zones.
- Each **Availability Zone** is a fully isolated partition of the AWS infrastructure.
 - Availability Zones consist of discrete **data centers**
 - They are designed for fault isolation
 - They are interconnected with other Availability Zones by using high-speed private networking
 - You choose your Availability Zones.
 - **AWS recommends replicating data and resources across Availability Zones** for resiliency.



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2. Availability Zone (AZ)

- **Definition** : An Availability Zone is a physically separate data center within an AWS Region.

- **Purpose** : AZs are designed to provide fault tolerance and high availability for applications. They ensure that if one AZ fails, others in the same Region can continue operating.
- **Characteristics** :
 - Each AZ is located at a significant distance from other AZs within the same Region to minimize the risk of simultaneous failures.
 - AZs are connected via high-bandwidth, low-latency networks.
 - A Region typically has at least three AZs (though some may have fewer or more).
 - Resources can be distributed across multiple AZs to achieve redundancy and improve reliability.
- **Use Case** : To build highly available and fault-tolerant systems, you can deploy your resources (e.g., EC2 instances, RDS databases) across multiple AZs within the same Region.

AWS data centers

- AWS data centers are **designed for security**.
- Data centers are where the data resides and data processing occurs.
- Each data center has redundant power, networking, and connectivity, and is housed in a separate facility.
- A data center typically has 50,000 to 80,000 physical servers.



Points of Presence

- AWS provides a global network of **Points of Presence** locations
- Consists of **edge locations** and a much smaller number of **Regional edge caches**
- Used with Amazon CloudFront
 - A global Content Delivery Network (CDN), that delivers content to end users with **reduced latency**
- Regional edge caches used for content with infrequent access.



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3. Edge Location

- **Definition** : An Edge Location is a point of presence (PoP) used by AWS services like Amazon CloudFront for caching content closer to end users.
- **Purpose** : Edge Locations reduce latency by serving cached content directly from locations near the user, rather than fetching it from the originating Region.
- **Characteristics** :
 - Edge Locations are not full-fledged data centers but rather smaller facilities optimized for caching and content delivery.
 - They are part of the AWS Global Accelerator and CloudFront network.
 - Edge Locations are spread across many more cities than Regions or AZs.
 - They do not host user-managed resources like EC2 instances or RDS databases; they only cache static content or route traffic.
- **Use Case** : If you want to deliver content quickly to users worldwide, you can use CloudFront to cache your data at Edge Locations.

4. Local Zones

- **Definition:**

A Local Zone is an AWS extension that places compute, storage, and other services closer to major cities or industry hubs for low-latency applications.

- **Purpose:**

It lets users run workloads that require single-digit millisecond latency near end users, while staying connected to the parent AWS Region.

- **Characteristics:**

- Linked to a specific AWS Region.
- Offers core services like EC2, EBS, and VPC locally.
- Reduces latency for real-time apps (e.g., gaming, streaming).
- Uses high-speed links to the parent Region.

- **Use Case:**

Ideal for organizations needing faster response times in cities far from main AWS Regions.

5. Wavelength

- **Definition:**

AWS Wavelength brings AWS compute and storage to the edge of 5G networks, hosted inside telecom providers' data centers.

- **Purpose:**

It enables ultra-low-latency mobile applications by processing data near users and devices instead of remote Regions.

- **Characteristics:**

- Deployed within 5G carriers' infrastructure.
- Keeps data traffic inside the telecom network.
- Offers familiar AWS services (like EC2 and EBS).
- Connects seamlessly to the parent Region.

- **Use Case:**

Perfect for 5G-based apps like AR/VR, IoT, and connected vehicles needing real-time responsiveness.

AWS infrastructure features

- **Elasticity and scalability**

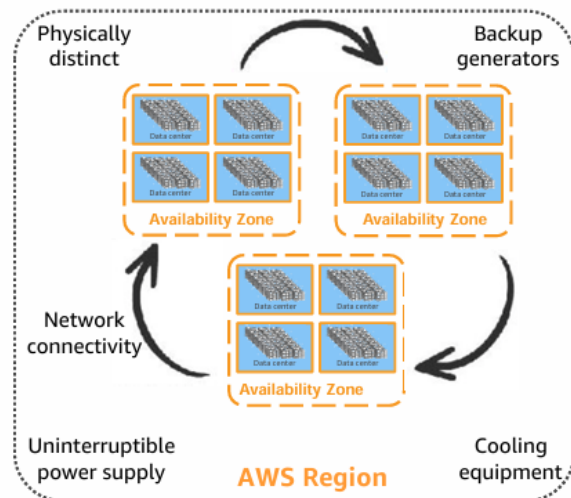
- Elastic infrastructure; dynamic adaption of capacity
- Scalable infrastructure; adapts to accommodate growth

- **Fault-tolerance**

- Continues operating properly in the presence of a failure
- Built-in redundancy of components

- **High availability**

- High level of operational performance
- Minimized downtime
- No human intervention



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Key takeaways



- The **AWS Global Infrastructure** consists of **Regions** and **Availability Zones**.
- Your choice of a **Region** is typically based on **compliance requirements** or to **reduce latency**.
- Each **Availability Zone** is physically separate from other Availability Zones and has redundant power, networking, and connectivity.
- **Edge locations**, and **Regional edge caches** improve performance by **caching** content closer to users.



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Note:

Amazon CloudFront is a **content delivery network (CDN)** service offered by AWS. It allows you to deliver content, such as web pages, images, videos, and APIs, to users with low latency and high transfer speeds by caching content at globally distributed edge locations. By leveraging CloudFront, you can improve the performance of your applications and reduce the load on your origin servers.

